Unit

28 Addition polymers

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Monomers, polymers and repeating units
- Addition polymerization
- Some common addition polymers

28.1 Plastic items in modern homes

Today plastics play a very important part in our modern-day lives. The use of plastics is increasing all the time as they replace materials such as metal, wood, paper and glass in a wide variety of uses (Fig. 28.1).

In recent years, plastics are used extensively in electrical components and interior and exterior furnishing of automobiles (Fig. 28.2). The automotive industry chooses plastics for their durability, corrosion resistance, ease of colouring and finishing, and lightweight, in addition to their favourable cost. The weight reduction achieved through the use of plastics becomes apparent in improved acceleration and lower fuel consumption.



Fig. 28.1a Plastic bucket and metal bucket



Fig. 28.1b Wooden, plastic and steel rulers



Fig. 28.1c Paper grocery bag and plastic grocery bag



Fig. 28.2 The plastic parts of a car — bumper (left) and dashboard (right)

Do you know

Birth of the plastics industry

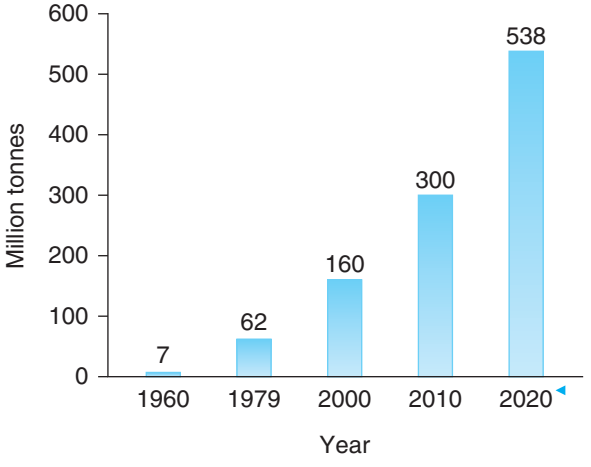
Until the middle of the nineteenth century, virtually all objects used in everyday life were made from naturally occurring or naturally derived materials.

In 1863, a billiard ball manufacturing company offered a prize to anyone who could find a substitute for natural ivory for making billiard balls.

An American named Hyatt and his brother figured out how to form billiard balls out of *cellulose nitrate* (made by modifying cellulose with nitric acid).

By 1871, Hyatt had established two companies to work with this new material, and the plastics industry was born.

The worldwide consumption of plastics has risen rapidly over the last 60 years (Fig. 28.3).



The figure for 2020 is an estimation.

Fig. 28.3 Worldwide consumption of plastics from 1960 to 2020

28.2 Why are plastics so useful?

Discussion

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website.*

Plastics are very useful materials. Many objects previously made with metals are now made with plastics. For each of the following objects, suggest advantages of using plastics over using metals in making the object.

1 Lamp posts

2 Door lock components

3 Helmets for soldiers

Discussion
1 • Electricity leakage can be prevented.
• Plastic does not corrode.
• Self-coloured / no painting required.
2 • Plastic does not corrode.
• No lubrication required.
• Quieter when slammed.
• Hard-wearing.
3 • Lightweight.
• Plastic does not corrode.
• No sharp edges.

cellulose nitrate 硝化纖維

In general, plastics are

- usually strong;
- lightweight;
- hard-wearing;
- good insulators of heat and electricity;
- usually flexible;
- waterproof;
- can be moulded into any shape easily;
- can be coloured easily;
- do not corrode;
- unreactive and inert to most substances;
- cost-effective.

28.3 What is a polymer?

Plastics are **polymers**.

✓ A polymer is a compound which consists of very large molecules formed by joining together many small molecules repeatedly.

Look at the separate paper clips in Fig. 28.4. They are single units. Then look at the chains of paper clips in Fig. 28.5. Each chain is made up of many paper clips joined together.



Fig. 28.4 Separate paper clips

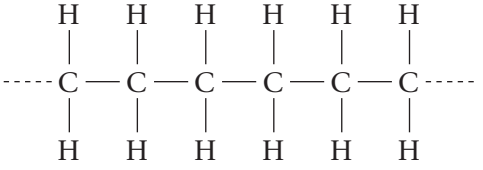


Fig. 28.5 Many paper clips joined together to form chains

polymer 聚合物

A polymer molecule is also a long-chain molecule. It is made up of many **monomer** molecules joined together. The process is called **polymerization**.

For example, many ethene (monomer) molecules polymerize to form one polythene (polymer) molecule.



part of a polythene molecule

Natural and man-made polymers

Many polymers occur naturally. They are **natural polymers**. Examples include starch, wool and silk (Fig. 28.6).

There are many **man-made polymers** as well. Man-made polymers include **regenerated polymers** (made by modifying natural polymers) and **synthetic polymers** (made from small molecules of monomers). These man-made polymers are known as plastics (Fig. 28.7).



Fig. 28.6 Silk, a natural polymer, has long been popular for high-quality clothing



Fig. 28.7 These plastic crates are made of man-made polymers

monomer 單體　polymerization 聚合作用　natural polymer 天然聚合物　man-made polymer 人造聚合物
regenerated polymer 再造聚合物　synthetic polymer 合成聚合物

Rayon is obtained by modifying cellulose.

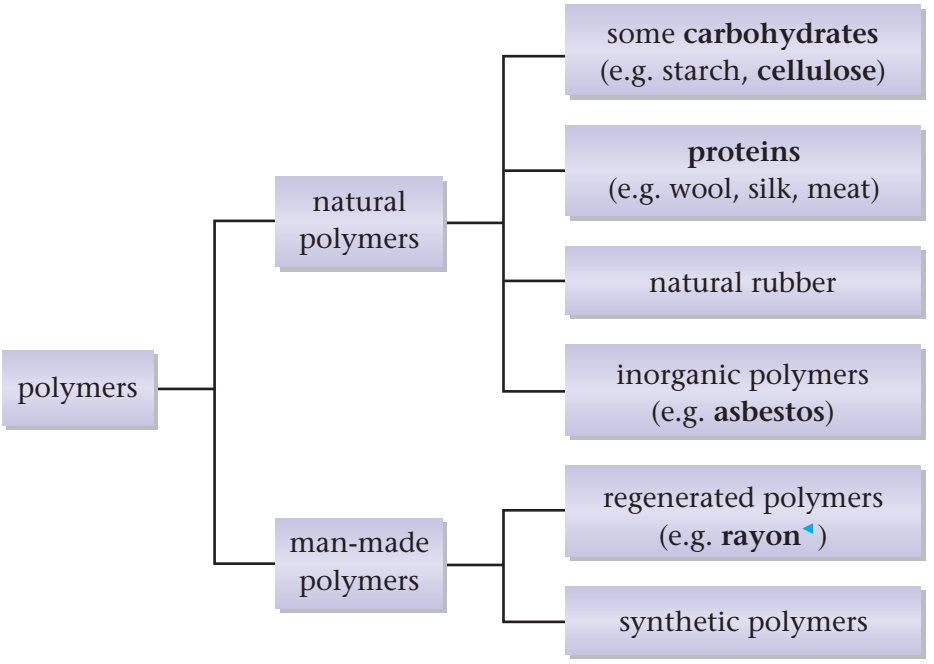


Fig. 28.8 Classification of polymers

Reference Website

Refer to a cartoon showing the processes involved in the production of plastics from petroleum:
http://www.totalpetrochemicalsusa.com/flash/WellheadToPlastics.swf
(accessed August 2014)

28.4 Polymerization of ethene

Alkenes are very useful products obtained from the cracking of alkanes. For example, ethene (C_2H_4) is the most important substance for producing synthetic polymers.

Polythene may also be spelled as polyethene.

Under the right conditions, ethene molecules can join together to form very long polymer chains (Fig. 28.9). Part of the double bond of an ethene molecule is broken, and the electrons in it are used to join to neighbouring molecules. All the carbon atoms in a polythene molecule are connected by single bonds.

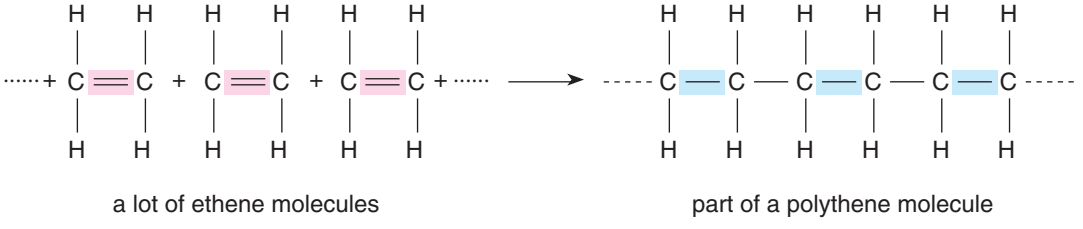


Fig. 28.9 Formation of a polythene molecule from ethene molecules

Reference Website

Refer to animations showing the polymerization of ethene:
- http://www.absorblearning.com/media/item.action?quick=11s (accessed August 2014)
- http://www.knockhardy.org.uk/polymer.htm (accessed August 2014)
- http://www.dynamicscience.com.au/tester/solutions/chemistry/polyetheneld.htm (accessed August 2014)

carbohydrate 碳水化合物 cellulose 纖維素 protein 蛋白質 asbestos 石棉 rayon 人造絲

CE 2011 Paper 1 Q7(b)
DSE 2014 Paper 1B Q3(b)(ii)

This is an example of **addition polymerization**. N1

N1 Remind students to be careful with the term 'addition polymerization' NOT ✔ 'additional polymerization'. ✘

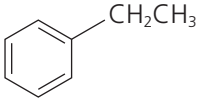
✓ Addition polymerization is a reaction in which monomer molecules join together repeatedly to form polymer molecules. No atoms are lost from the monomer molecules during the reaction.

Polymers that are made by addition polymerization are called **addition polymers**. Thus, polythene is an addition polymer.

Monomers that undergo addition polymerization usually contain carbon-carbon double bonds◀, for example, ethene and propene.

DSE 2012 Paper 1A Q21

◀ The C=C bonds in benzene and the phenyl group of aromatic compounds do NOT undergo addition polymerization.

For example, 

does NOT undergo addition polymerization.

DSE 2012 Paper 1A Q21

Do you know

The discovery of polythene

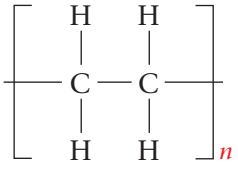
Polythene was discovered in 1933 by accident by two British chemists, Fawcett and Gibson, who worked for Imperial Chemical Industries (a large chemical company). They were trying to investigate the reaction of benzaldehyde and ethene gas at 1900 atm and 170 °C. When they added ethene gas, they let some air in. Few days later, they found a waxy solid inside the reaction vessel. They decided that the ethene gas must have reacted with themselves to form the solid and they worked out that the solid was polythene. Later, further studies concluded that the polymerization reaction of ethene requires a suitable amount of oxygen to initiate the process.

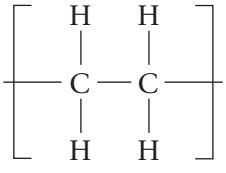
The British Telegraphic Construction and Maintenance Company recognized that the new material was a good insulator for underwater telegraph or telephone cables. In 1939, enough polythene was made for one nautical mile of underwater cable.

What is a repeating unit?

Polymers have no fixed size. For example, the number of ethene molecules in a polythene chain may range from 1 000 to 100 000. We can represent the structure of polythene by the following general formula:

CE 2009 Paper 1 Q4(b)(i)
CE 2011 Paper 1 Q7(a)



, where 

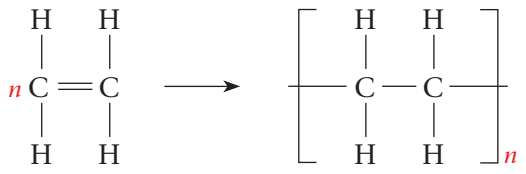
is the **repeating unit**

and n is the number of units present in the polythene chain.

addition polymerization 加成聚合作用　　addition polymer 加成聚合物　　repeating unit 重複單位

✓ A repeating unit is the smallest part of a polymer molecule, and the whole polymer structure can be obtained by repeating it.

We can also represent the polymerization of ethene in the following way:



CE 2006 Paper 1 Q11(b)(i)
CE 2009 Paper 1 Q4(b)(i)

Low density polythene and high density polythene

Different forms of polythene can be made depending on the conditions of temperature, pressure and catalyst. **Low density polythene** (LDPE) contains relatively short chains with a certain amount of branching (Fig. 28.11a). The branched chains do not pack together well. Thus, low density polythene is quite soft, flexible and has a fairly low density. These properties make it suitable for making grocery bags and sheeting.

High density polythene (HDPE) contains much longer chains with much less branching. The chains can pack together well (Fig. 28.11b). This makes the plastic stiffer, stronger and has a higher density. High density polythene is used to make bottles and buckets for which low density polythene would be insufficiently strong. N2

N2 Comparing low density polythene and high density polythene

	Low density polythene (LDPE)	High density polythene (HDPE)
Degree of branching from chain	high	low
Density (g cm^{-3})	0.91 – 0.94	0.95 – 0.97
Tensile strength (MPa)	5 – 15	20 – 40
Melting point (°C)	~115	~135

Tensile strength measures the ability of a material to withstand a stretching force. The unit used is megapascal MPa (where 1 MPa = 10^6 N m^{-2}).

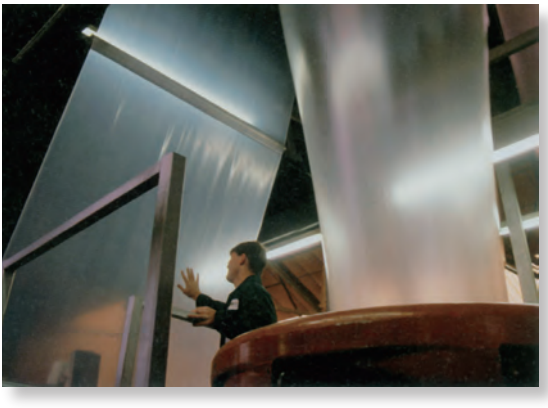


Fig. 28.10 Production of LDPE film

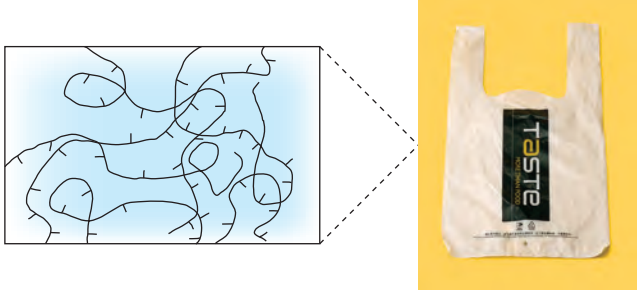


Fig. 28.11a Structure of LDPE

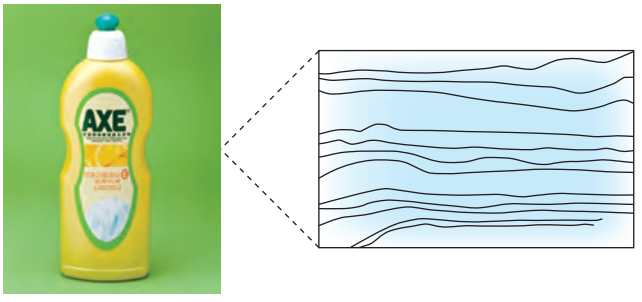


Fig. 28.11b Structure of HDPE

low density polythene 低密度聚乙烯　　high density polythene 高密度聚乙烯

Practice 28.1 Refer to page T42.

Polythene can be obtained from petroleum in three stages.

petroleum —*Stage 1*→ naphtha —*Stage 2*→ ethene —*Stage 3*→ polythene

a) Name the process for obtaining naphtha from petroleum in *Stage 1*.

b) What are the TWO main processes involved for obtaining ethene from naphtha in *Stage 2*? DSE 2013 Paper 1B Q6

c) i) Draw the repeating unit of polythene.

ii) Write a chemical equation for the formation of polythene from ethene.

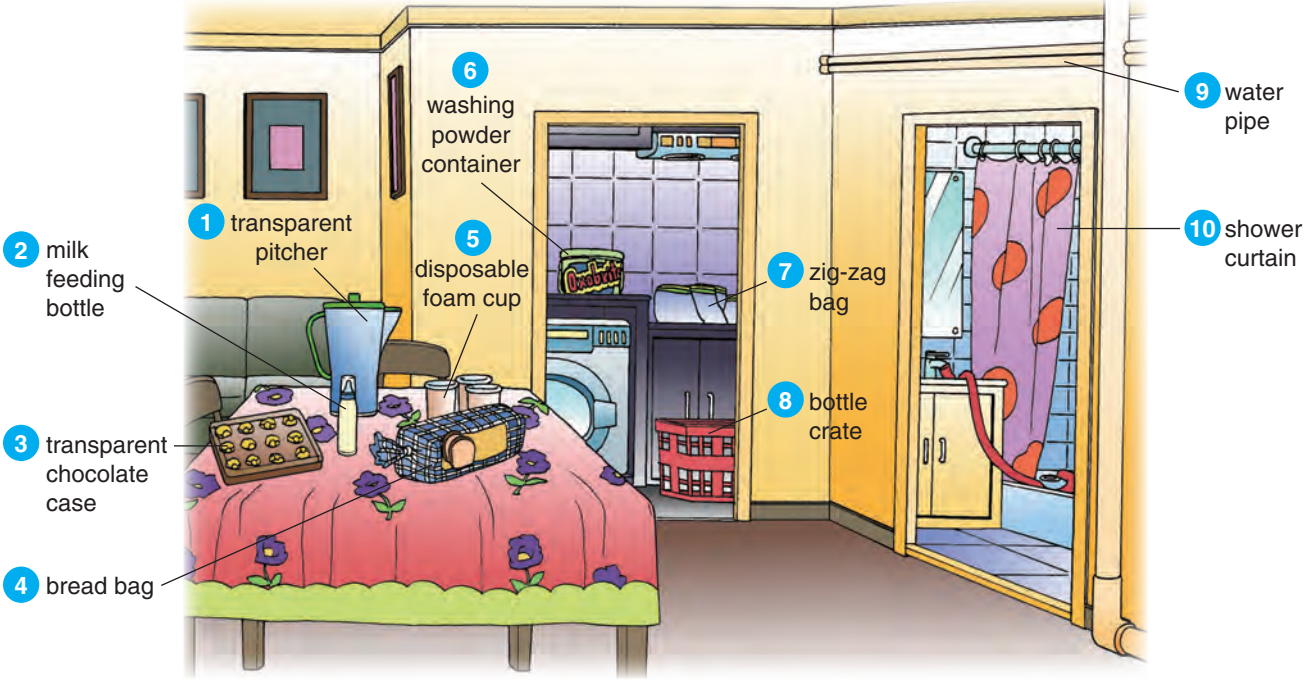
iii) Name the type of polymerization involved in the production of polythene.

28.5 Some common addition polymers

Discussion

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Ten products made of addition polymers are shown in the figure below.



What properties should a polymer have so that it could be used in each case? Name TWO properties for each usage. The first one has been done as an example.

Product	Two properties the polymer used should have
1 Transparent pitcher	• waterproof • non-toxic

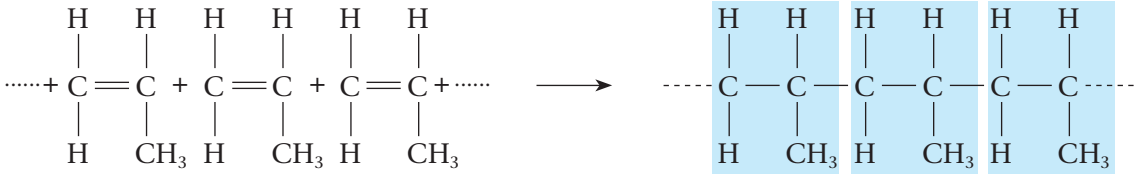
Refer to page T42.

You may not be able to name an addition polymer suitable for manufacturing each product in the above figure. To do that, you need to study various addition polymers and their properties.

We can produce other monomers by substituting the hydrogen atoms in ethene with different atoms or groups of atoms. These monomers can be polymerized to make addition polymers with different properties. Let us look at some common addition polymers. N3

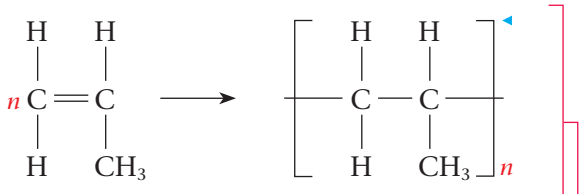
N3 Refer to page T44.

Polypropene (PP)



propene polypropene

or

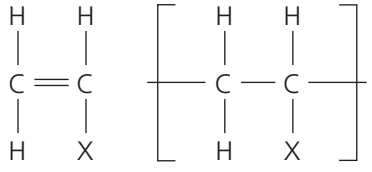


propene **polypropene**

CE 2010 Paper 1 Q12(f)

To deduce the repeating unit of a polymer from a given monomer with a C=C bond, simply turn the double bond into a C–C single bond and show the bonds on the other sides of the two C atoms that would continue the chain.

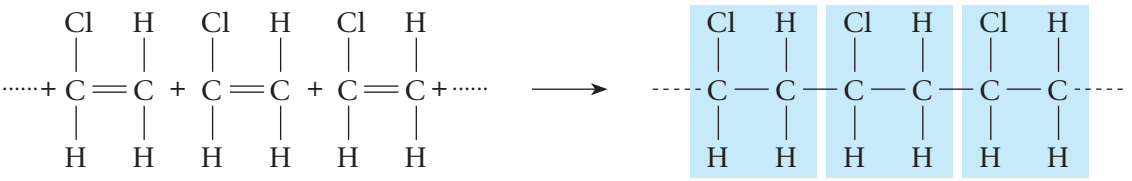
For example,



monomer repeating unit

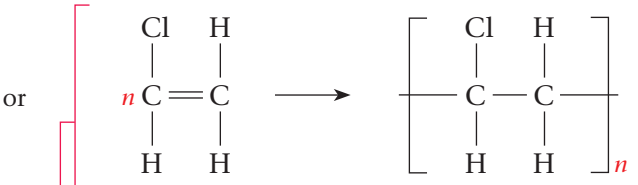
polypropene 聚丙烯

Polyvinyl chloride (PVC)



vinyl chloride — polyvinyl chloride

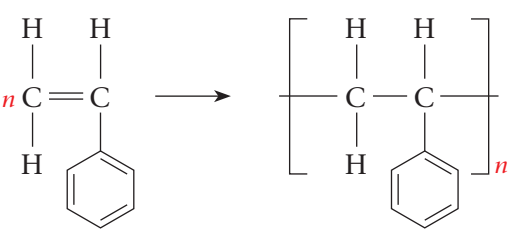
or



vinyl chloride — **polyvinyl chloride**

CE 2011 Paper 1 Q7(e)(i)

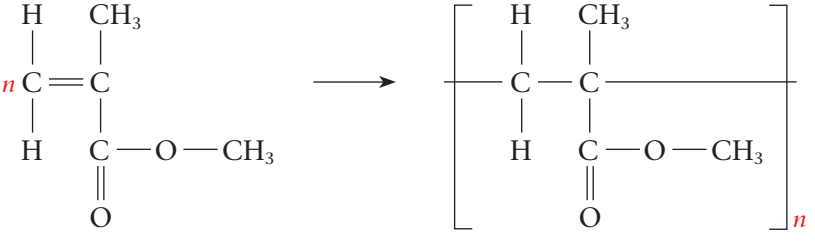
Polystyrene (PS)



styrene — **polystyrene**

ASL 2011 Paper 2 Q7(c)(i)

Perspex



methyl 2-methylpropenoate — **Perspex**
poly(methyl 2-methylpropenoate)

Poly(methyl 2-methylpropenoate) is often known as Perspex, one of its trade names.

All the addition polymers described here are **thermoplastics**. Thermoplastics can be melted or softened by heat at relatively low temperatures.

CE 2008 Paper 2 Q25

Video
Investigating the properties of some polymers

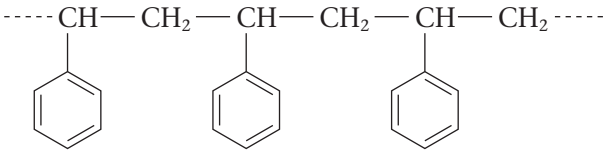
28.1
Investigating the properties of some polymers.

28.2
Preparing polystyrene.

polyvinyl chloride 聚氯乙烯　polystyrene 聚苯乙烯　Perspex 有機玻璃　thermoplastic 熱塑性塑膠

Example 28.1

Q *Expanded polystyrene* is commonly used to make disposable lunch boxes. Polystyrene has the following structure:



a) i) Draw the structure of styrene, the monomer of polystyrene.

ii) What characteristic in the structure of the monomer enables it to act as a monomer?

b) Suggest why polystyrene does NOT have a constant relative molecular mass.

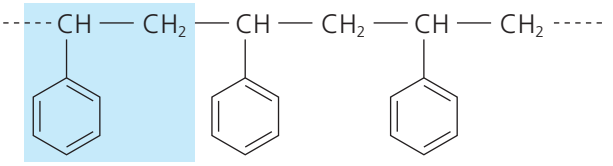
c) Expanded polystyrene can be used to make foam cups and lunch boxes.

i) Suggest ONE reason why polystyrene is expanded before it is used to make cups and lunch boxes.

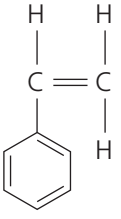
ii) Suggest ONE other common use of expanded polystyrene. State ONE property of expanded polystyrene associated with this usage.

It is possible to identify the monomer from which the polymer has been formed from the structure of the polymer.

For example, the structure of polystyrene is shown below with its repeating unit shaded.

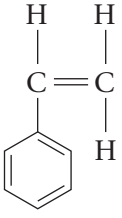


The monomer is based on this shaded region and must include a C=C bond.



monomer

A a) i)



ii) It contains carbon-carbon double bond.

b) Polystyrene is a mixture of polymeric molecules of different chain lengths.

c) i) Expanded polystyrene is a good insulator of heat.

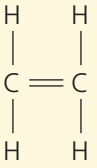
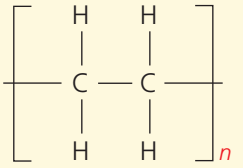
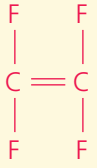
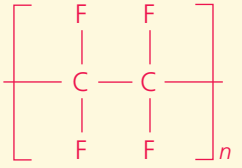
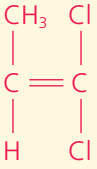
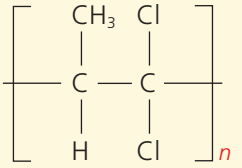
ii) Packaging material

It is light but still quite rigid. / It is an excellent shock absorbent.

expanded polystyrene 發泡聚苯乙烯

Practice 28.2

1 The following table shows some monomers and polymers. Complete the table. The first one has been done as an example.

IUPAC name of monomer	Structure of monomer	Structure of polymer
ethene		
tetrafluoroethene		
1,1-dichloropropene		

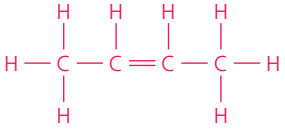
2 Consider but-2-ene.

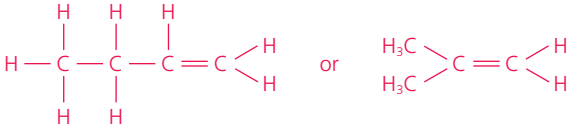
a) Write the structural formula of but-2-ene.

b) Alkene X has the same molecular formula as but-2-ene. Draw a possible structure of X.

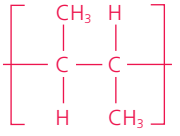
c) Both but-2-ene and alkene X undergo addition polymerization to give polymers. Draw the repeating unit of each polymer formed.

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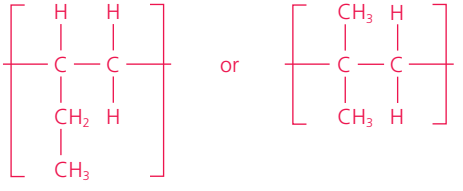
2 a) 

b) 

c) But-2-ene:



Alkene X:



N4 About 85% of the world plastics consumption is from just four polymers: polythene, polypropene, polystyrene and polyvinyl chloride.

N5 Refer to page T45.

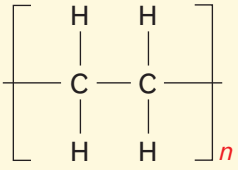
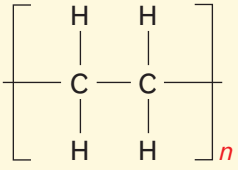
28.6 Uses of some common addition polymers

Table 28.1 shows the uses and properties of the above common addition polymers. N4 N5

A flexible material is easily deformed without breaking when a force is applied.

Table 28.1

Uses and properties of some common addition polymers

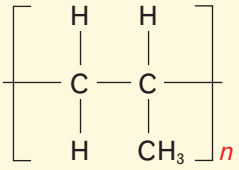
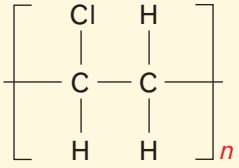
Name	Structure of polymer	Properties of polymer	Uses of polymer	Products
Low density polythene (LDPE)		• lightweight • low-melting • soft and flexible • good resistance to chemical attack • good moisture barrier properties • very good electrical insulator	• films and sheets for packaging • wrapping • grocery bags • squeezable bottles	 
High density polythene (HDPE)		• hard to semi-flexible • stiffer and stronger than low density polythene • excellent chemical resistance • excellent moisture barrier properties	• milk, water and juice containers • buckets • washing-up bowls • grocery bags • toys	 

Continued on next page

A hard material is difficult to indent or scratch.

A stiff material can resist being deformed when a force is applied.

A strong material can withstand a large force before it breaks.

Name	Structure of polymer	Properties of polymer	Uses of polymer	Products
Polypropene (PP)		• stronger than high N6 density polythene • high melting point • very low density • excellent chemical resistance	• margarine tubs • medicine bottles • storage boxes • crates • ketchup bottles • car bumpers N7	  
Polyvinyl chloride (PVC)		PVC • stiff • brittle • excellent chemical resistance • good weathering ability PVC with *plasticizer* N8 • more flexible	• drain pipes • bottles • floor tiles • shower curtains and raincoats • insulating tapes and coverings round electrical wires • as 'artificial leather' in making seat coverings for taxis ASL 2006 Q13(b)(I)(i) N9	  

N6 Refer to page T45.

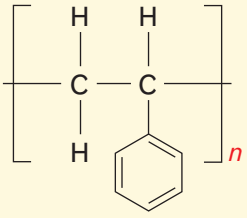
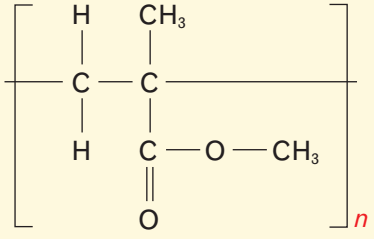
Continued on next page

N7 Polypropene can be used to make surgical face masks.

N8 Refer to page T46.

N9 PVC can be used to make food packing material. However, it may release monomer of PVC / short chain molecule of PVC / plasticizer that can contaminate the food.

plasticizer 增塑劑

Name	Structure of polymer	Properties of polymer	Uses of polymer	Products
Polystyrene (PS)		Polystyrene • transparent • hard and rigid • brittle Expanded polystyrene◀ • white • extremely light solid foam • excellent shock absorbent	• 'see-through' containers such as sample bottles, cups, etc. • compact disc cases • yogurt pots • disposable cups and lunch boxes • packing materials for delicate articles (e.g. TV or computer)	  CE 2007 Paper 2 Q9
Perspex		• highly transparent • strong • rigid • not easily scratched	• Glass substitute for advertizing displays and illuminating signs • protective shields • car light covers • aircraft windows	 

◀ Expanded polystyrene is made by adding a blowing agent to polystyrene. Air is trapped in expanded polystyrene and this enhances its heat insulating property.

Reference Website

- Refer to an animation showing injection moulding: http://www.bpf.co.uk/Data/Image/InjectionMouldingAnimation.swf (accessed August 2014)
- Refer to an animation showing extrusion blow moulding: http://www.bpf.co.uk/Data/Image/Extrusion%20Blow%20Moulding.swf (accessed August 2014)

Worksheet

A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

Now you have studied the properties of some common addition polymers. Return to the discussion in Section 28.5.

Suggest a polymer suitable for manufacturing each product. Explain your choice in each case. The first one has been done as an example.

Product	Polymer suitable for manufacturing the product	Properties of the chosen polymer that make it suitable
1 Transparent pitcher	Perspex	Perspex is highly transparent, waterproof, rigid, easily shaped, non-toxic and chemically inert.

Refer to page T47.

Example 28.2

Q The table shows the properties of two polymers.

Polymer	Strength	Flexibility	Melting point (°C)
Low density polythene	weak	very flexible	90
High density polythene	strong	stiff	140

a) High density polythene is well known for its resistance to corrosion by water, acids, alkalis and oxidizing agents.

Explain why it is resistant to such chemical attacks.

b) High density polythene is better than low density polythene for making plastic coffee cups for vending machines.

Suggest TWO advantages of using high density polythene in this application.

A a) High density polythene is a hydrocarbon with strong C–C and C–H bonds.

These bonds are not readily attacked by chemicals.

b) High density polythene is stiffer and melts at a higher temperature.

Practice 28.3

1 Polyvinyl chloride (PVC) can be used to make window frames. Consider the following advertisement.

WONDERFUL WINDOWS

Replace your old window frames with our super high quality plastic window frames!

NO PAINTING – MAINTENANCE FREE

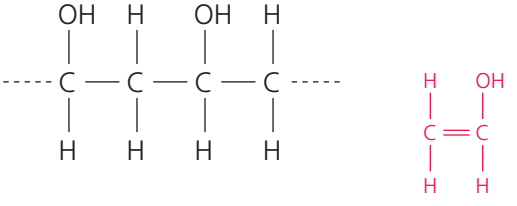
a) Explain why PVC window frames need no painting or maintenance.

Polyvinyl chloride does not corrode.

b) Explain whether PVC is suitable for making the handle of a frying pan.

Not suitable because PVC is a thermoplastic / softens upon heating.

2 Polyethenol is a very unusual polymer because it can dissolve in water under certain conditions.



a) Draw the structure of the monomer of polyethenol.

b) Suggest why polyethenol can dissolve in water.

The polymer contains many –OH groups. Thus, it can form hydrogen bonds with water, facilitating its water solubility.

Have you mastered?

Key terms

plastic 塑膠 105 polymer 聚合物 107 monomer 單體 108 polymerization 聚合作用 108

natural polymer 天然聚合物 108 man-made polymer 人造聚合物 108

regenerated polymer 再造聚合物 108 synthetic polymer 合成聚合物 108

carbohydrate 碳水化合物 109 cellulose 纖維素 109 protein 蛋白質 109 asbestos 石棉 109

rayon 人造絲 109 addition polymerization 加成聚合作用 110

addition polymer 加成聚合物 110 repeating unit 重複單位 110

low density polythene 低密度聚乙烯 111 high density polythene 高密度聚乙烯 111

polypropene 聚丙烯 113 polyvinyl chloride 聚氯乙烯 114 polystyrene 聚苯乙烯 114

Perspex 有機玻璃 114 thermoplastic 熱塑性塑膠 114 expanded polystyrene 發泡聚苯乙烯 115

plasticizer 增塑劑 118

Checklist

After studying this unit, you should be able to

- ☐ recognize that synthetic polymers are built up from small molecules called monomers;
- ☐ understand that alkenes and unsaturated carbon compounds can undergo addition polymerization;
- ☐ describe addition polymerization using chemical equations;
- ☐ describe the structural properties and uses of some common addition polymers:
  - polythene;
  - polypropene;
  - polyvinyl chloride;
  - polystyrene;
  - Perspex;
- ☐ deduce the repeating unit of an addition polymer obtained from a given monomer;

- ☐ deduce the structure of the monomer from a given section of the structure of an addition polymer;
- ☐ describe the effect of heat on thermoplastics.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

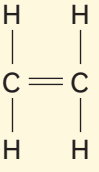
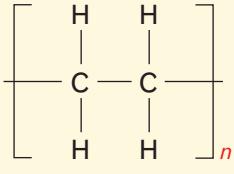
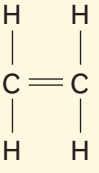
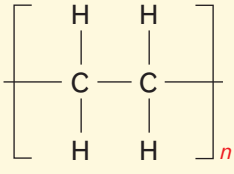
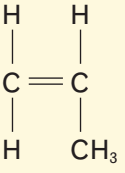
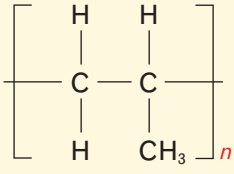
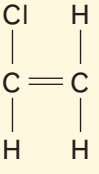
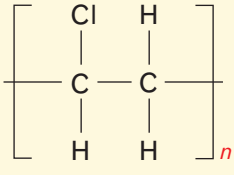
Summary

1 A polymer is a compound which consists of very large molecules formed by joining together many small molecules repeatedly.

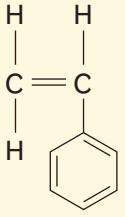
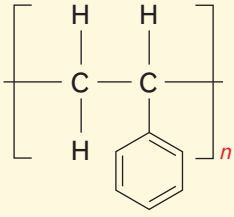
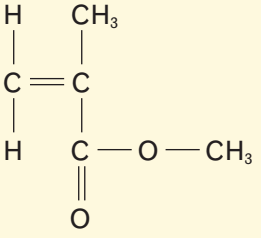
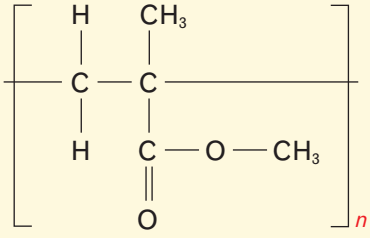
2 Addition polymerization is a reaction in which monomer molecules join together repeatedly to form polymer molecules. No atoms are lost from the monomer molecules during the reaction.

3 A repeating unit is the smallest part of a polymer molecule, and the whole polymer structure can be obtained by repeating it.

4 The information of some common addition polymers are shown below:

Polymer	Structure of monomer	Structure of polymer
Low density polythene (LDPE)		
High density polythene (HDPE)		
Polypropene (PP)		
Polyvinyl chloride (PVC)		

Continued on next page

Polymer	Structure of monomer	Structure of polymer
Polystyrene (PS)		
Perspex		

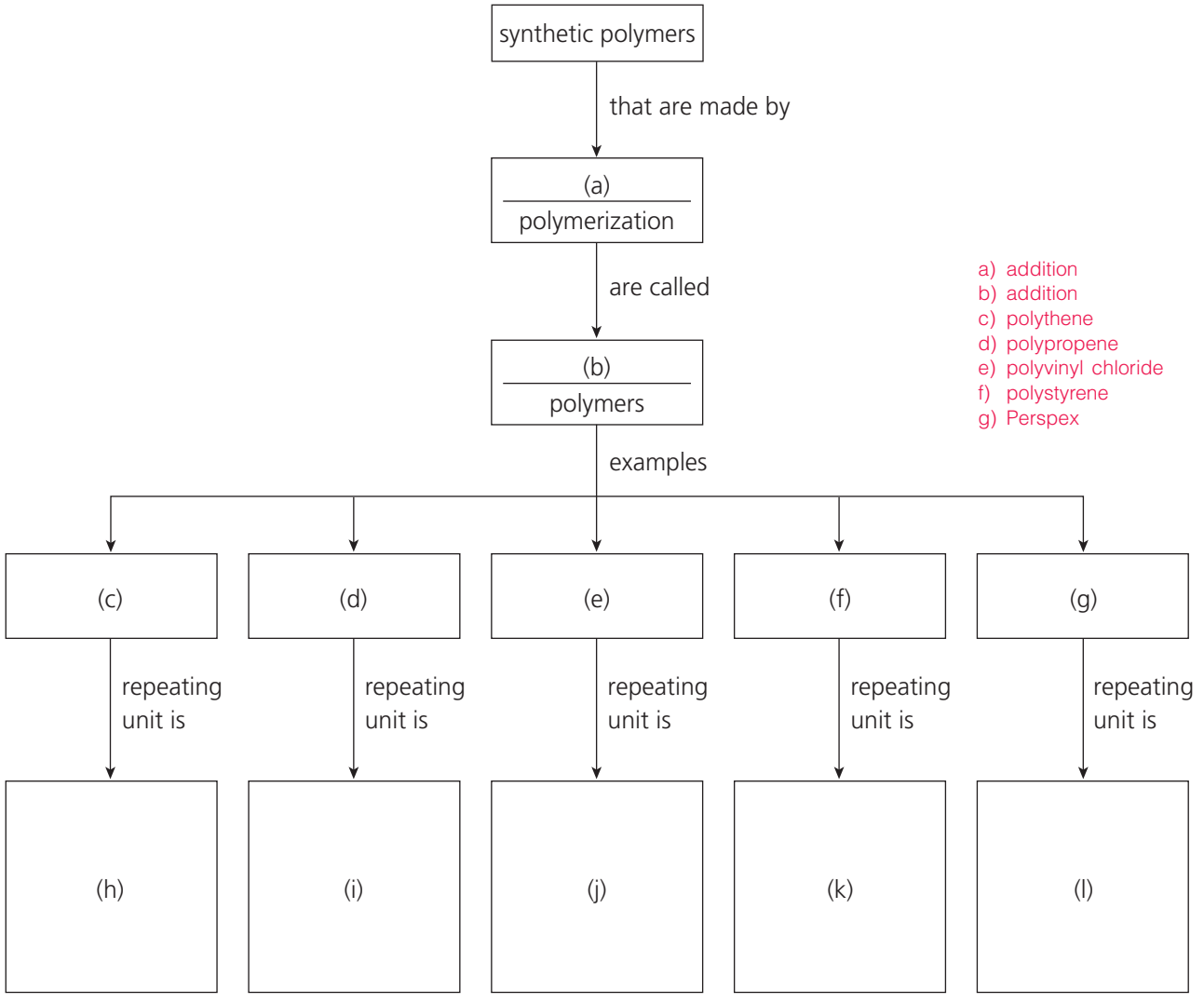
The above addition polymers are thermoplastics. They can be melted or softened by heat at relatively low temperatures.

Unit Exercise

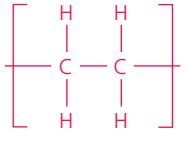
Note: The symbol ! indicates the level of difficulty of a question.

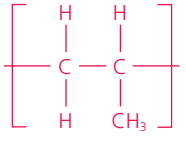
Part I Knowledge and understanding

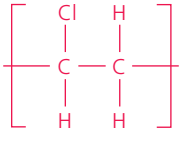
1 Complete the following concept map.

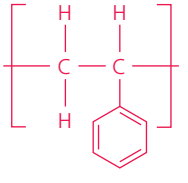


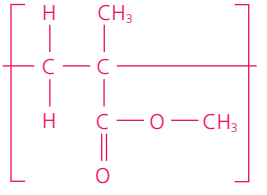
a) addition
b) addition
c) polythene
d) polypropene
e) polyvinyl chloride
f) polystyrene
g) Perspex

h) 

i) 

j) 

k) 

l) 

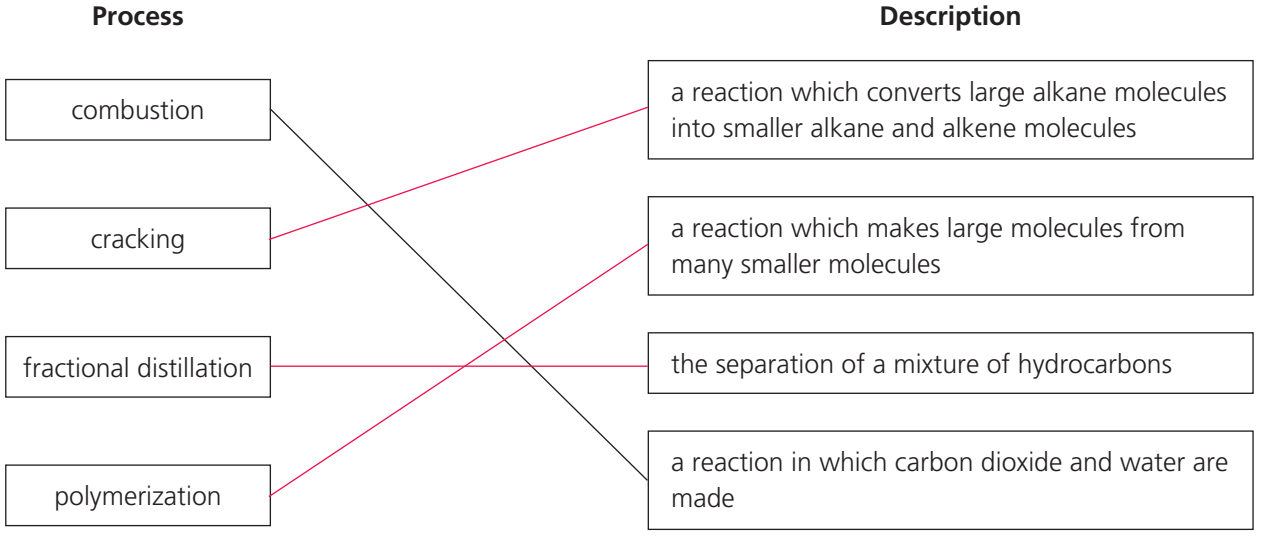
2 Explain the following terms:

a) polymers;
A polymer is a compound which consists of very large molecules formed by joining together many small molecules repeatedly.
b) addition polymerization; Addition polymerization is a reaction in which monomer molecules join together repeatedly to form polymer molecules. No atoms are lost from the monomer molecules during the reaction.
c) repeating unit.
A repeating unit is the smallest part of a polymer molecule, and the whole polymer structure can be obtained by repeating it.

3 Suggest TWO advantages and ONE disadvantage of using a synthetic polymer to replace the old material in each of the following applications.
Refer to page T49.

Object	Synthetic polymer now used	Old material used	Advantages of using synthetic polymer	Disadvantage of using synthetic polymer
Lenses				
Water pipes				
Bucket				
Packaging				
Disposable cups				

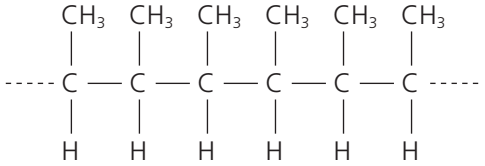
4 This question is about different types of chemical processes. Match the process with its correct description. One has been done for you.



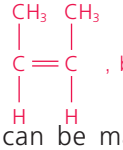
(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 1, Jan. 2010, 5(a))

Part II Multiple choice questions

5 The structure of polymer X can be represented by



What is the monomer of X?

A 1,1-dimethylethene
B 1,2-dimethylethene
C But-1-ene
(D) But-2-ene The monomer of X is  , but-2-ene.

6 Which of the following items can be made from polystyrene?

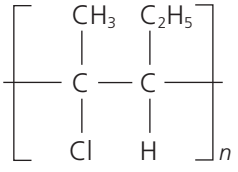
A Clothing
B Food wrap
C Electric socket
(D) Packaging material

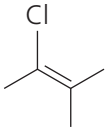
(HKCEE, Paper 2, 2007, 9)

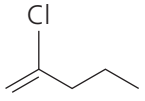
7 Which of the following combinations concerning the uses of synthetic polymers is correct?

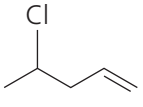
	Synthetic polymer	Use
A	low density polythene	bottle crates
(B)	polypropene	car bumpers
C	polyvinyl chloride	compact disc cases
D	perspex	raincoats

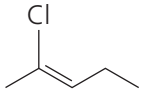
8 Which of the following monomers would form the polymer below?



A 

B 

C 

(D) 

(Edexcel Advanced GCE, Chemistry, Unit 5, Jan. 2011, 14)

9 Which of the following statements concerning thermoplastics are correct?

(1) They can be softened at relatively low temperatures.
(2) They are all highly transparent.
(3) They are inert to most substances.

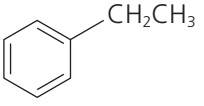
A (1) and (2) only
(B) (1) and (3) only (2) NOT all thermoplastics are transparent.
C (2) and (3) only
D (1), (2) and (3)

10 Which of the following statements concerning polyvinyl chloride (PVC) are correct?

(1) It softens upon heating.
(2) It is a thermoplastic.
(3) It is used in making shower curtains.

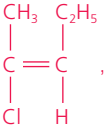
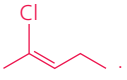
A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
(D) (1), (2) and (3)

11 Which of the following compounds can be used as monomers to make addition polymers?

(1) $CF_2=CF_2$
(2) $CH_2=C(CH_2CH_3)CN$
(3) 

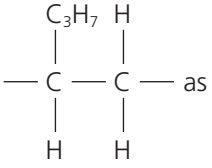
(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKDSE, Paper 1A, 2012, 21)

8 The monomer is  , i.e.  .

12 Which of the following statements about pent-1-ene is / are correct?

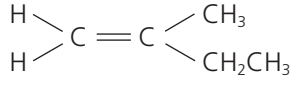
(1) It has the same molecular formula as methylbutane.

(2) It can form a polymer with  as the repeating unit.

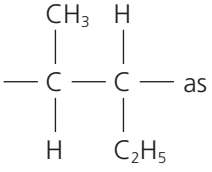
(3) It can decolourize acidified potassium dichromate solution.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

13 Consider the following carbon compound:
Refer to page T50 for explanation.

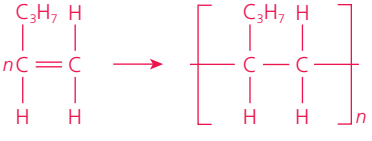


Which of the following statements about this compound is / are correct?

(1) Its IUPAC name is 1-ethyl-1-methylethene.
(2) It can decolourize bromine dissolved in an organic solvent.
(3) It can form a polymer with  as the repeating unit.

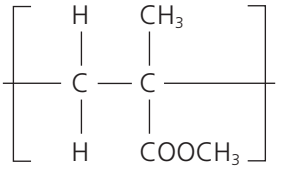
A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

12 (1) The molecular formula of pent-1-ene is C_5H_{10} while that of methylbutane is C_5H_{12}.
(2) Pent-1-ene undergoes addition polymerization as shown below:



(3) Pent-1-ene does NOT react with acidified potassium dichromate solution.

14 The repeating unit of polymer X is shown below:

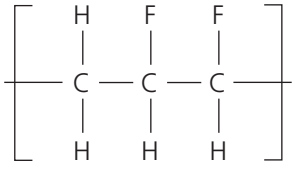


Which of the following statements about X is / are correct?

(1) X is an addition polymer.
(2) X is soft and flexible.
(3) X can be used to make car light covers.

A (1) only
B (2) only
C (1) and (3) only Polymer X is Perspex.
D (2) and (3) only (2) Perspex is strong and brittle.

15 A carbon compound X has the molecular formula $C_3H_4F_2$.
Refer to page T50 for explanation.
Which of the following statements about X is / are correct?

(1) It has at least four possible structures.
(2) It must be a saturated compound.
(3) It can form a polymer with  as the repeating unit.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

Part III Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

16 Chloroethene can be polymerized to form polychloroethene.

a) Write the structural formula of chloroethene.

b) Draw the repeating unit in polychloroethene.

c) What is the name given to this type of polymerization?

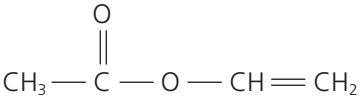
d) Use chloroethene and polychloroethene to explain the difference between the terms saturated and unsaturated.

e) Write the structural formula for the product of the reaction between chloroethene and chlorine.

(Edexcel GCE O Level, Paper 1, May 2009, 13(b))

17 Vinyl acetate is the monomer for the preparation of polyvinylacetate (PVA) which is widely used in the building industry.

Vinyl acetate has the structural formula:

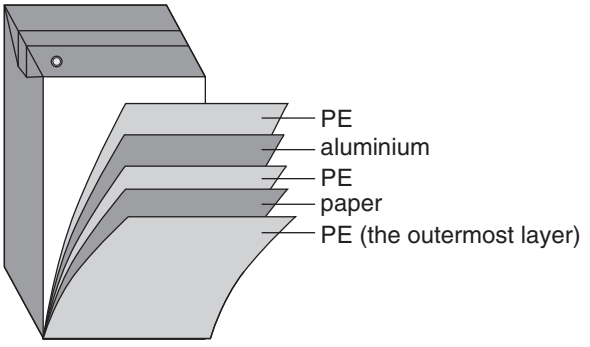


a) Draw part of the structure of polyvinylacetate, showing THREE monomer units joined together.

b) Name the type of polymerization involved in the production of polyvinylacetate.

c) What characteristic in the structure of vinyl acetate enables it to act as a monomer?

18 The diagram below shows the structure of a common beverage box consisting of layers of paper, polythene (PE) and aluminium.



a) Draw the repeating unit of PE.

b) Name the type of polymerization involved in making PE.

c) Explain the function of the outermost PE layer of the beverage box.

d) Oxygen can pass through paper and PE. Explain how the box can prevent the beverage from spoilage.

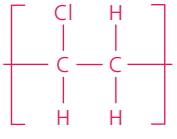
e) Polychloroethene, commonly called polyvinyl chloride (PVC), is also a polymer.

i) Draw the structure of the monomer of PVC.

ii) PVC can be used to make food packing material. However, it may release some substances to contaminate the food. Suggest ONE substance that may be released.

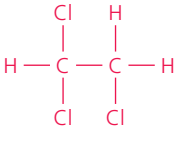
(HKCEE, Paper 1, 2011, 7)

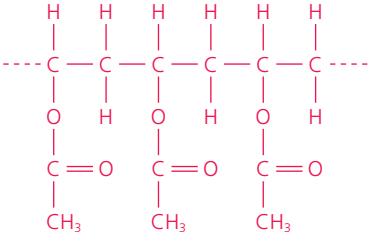
16 a) 

b) 

c) Addition polymerization

d) Chloroethene is unsaturated because it contains carbon-carbon double bond.
Polychloroethene is saturated because all the carbon atoms are connected by single bonds.

e) 

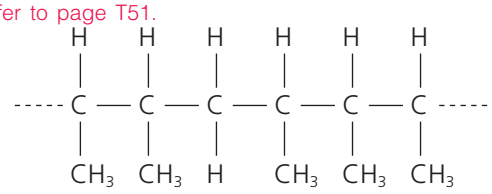
17 a) 

b) Addition polymerization

c) Vinyl acetate contains carbon-carbon double bond.

19 Co-polymers have a variety of uses. The diagram below shows a section of co-polymer A, which is produced from but-2-ene and another alkene monomer.

Refer to page T51.



co-polymer A

a) i) Name the type of polymerization that produces co-polymer A.

ii) Give the name and structural formula of the other alkene monomer that is reacted with but-2-ene to produce co-polymer A.

b) The monomers used to make co-polymer A are both alkenes. Describe a simple chemical test for an alkene.

(OCR Advanced Subsidiary GCE, Chem. B (Salters), F332, Jan. 2010, 3(a)–(b))

20 This question is about polymers. Polymers are made by polymerization. Small molecules join together to make a polymer.

a) Look at the table. It shows some information about polymers.

Polymer	Density (g cm^{-3})	Melting point (°C)	Solubility in oil
A	0.92	85	insoluble below 80 °C; soluble above 80 °C
B	0.96	120	insoluble below 80 °C; soluble above 80 °C
C	1.05	65	soluble
D	1.39	60	soluble
E	0.90	150	insoluble

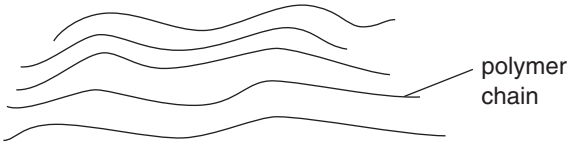
Which polymer would be best for making a pipe to carry oil at 100 °C?

Choose A, B, C, D or E.

Explain your answer.

E. It is insoluble in oil.
Its melting point is above 100 °C.

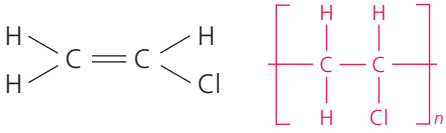
b) Look at the structure of polymer D.



Polymer D has a low melting point and can be stretched easily.

Explain why. The intermolecular forces between the polymer chains are weak.
The chains can slide past each other easily.
Use ideas about its structure.

c) Look at the structural formula of chloroethene.



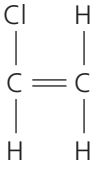
Chloroethene is used to make a polymer.

Write the structural formula of this polymer.

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 1, Jan. 2012, 1)

21 Chloroethene has the structure shown below:

Refer to page T51.



It forms polyvinyl chloride (PVC) by addition polymerization.

a) Explain the meaning of the term 'addition polymerization'.

b) Draw the repeating unit of PVC.

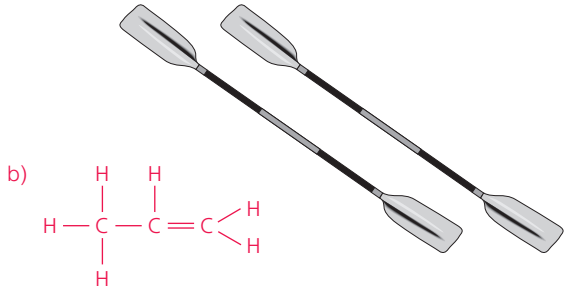
c) Products made of PVC may vary greatly in rigidity.

i) Give ONE flexible product made of PVC.

ii) Give ONE rigid product made of PVC.

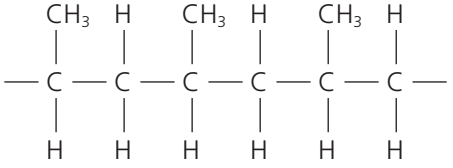
iii) Explain whether PVC is suitable for making the casing of a hair blower.

22 Polypropene is an addition polymer with many uses. Canoe paddles are often made of polypropene.



b)

Polypropene is formed from propene, C_3H_6. A section of a polypropene chain which is formed from three propene molecules is shown.



a) What is a polymer?

A polymer is a compound which consists of very large molecules formed by joining together many small molecules repeatedly.

b) Draw the structure of a propene molecule, C_3H_6, showing all covalent bonds.

c) In terms of the covalent bonds, explain how molecules of propene combine to form a polypropene molecule.

(Edexcel GCSE (Higher Tier), Chemistry, Unit C2, Jun. 2009, 1)

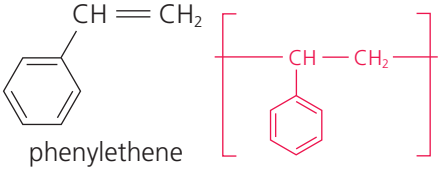
22 c) Part of the carbon-carbon double bonds of a propene molecule is broken,
and the electrons in it are used to join the neighbouring molecules.

23 Scientists have investigated the use of 'expanded' poly(phenylethene) for building houses in countries with cold climates and countries prone to earthquakes. 'Expanded' poly(phenylethene) is an insoluble polymer with a honeycomb structure containing air.

a) Suggest a property of 'expanded' poly(phenylethene), other than its insolubility, which makes it suitable for building houses in these countries.

Good thermal insulator / low density

b) The poly(phenylethene) monomer has the structure shown below.



phenylethene

Draw the structure of the repeating unit of poly(phenylethene).

c) Poly(phenylethene) is a thermoplastic.

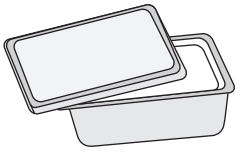
Explain the meaning of the term 'thermoplastic'.

(OCR Advanced Subsidiary GCE, Chem. B (Salters), F332, Jan. 2011, 2(a)–(c))

23 c) A plastic that can be melted or softened by heat at relatively low temperatures.

24 The following diagrams show three uses of synthetic polymers. These polymers are produced by addition polymerization.

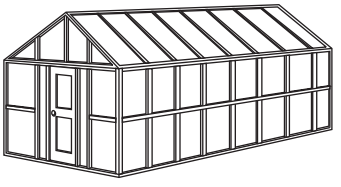
Refer to page T52.



making a margarine tub



as the inside coating of a non-stick frying pan



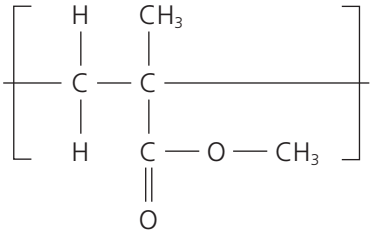
making windows of a greenhouse

a) Explain the meaning of the term 'addition polymerization'.

b) i) Suggest ONE synthetic polymer which is suitable for making the margarine tub.

ii) Give TWO properties of the suggested polymer which make it suitable for this use.

c) Perspex can be used to make the windows of greenhouses. The repeating unit of Perspex is shown below.



i) What is the meaning of 'a repeating unit'?

ii) Draw the structure of the monomer used in producing Perspex.

iii) One of the properties that makes Perspex suitable for making windows of a greenhouse is that it is transparent.

Suggest TWO other properties of Perspex which make it suitable for this use.

d) Teflon can be used as the inside coating of a non-stick frying pan. Teflon is a polymer of linear structure consisting of carbon and fluorine only. The ratio of number of carbon atoms : the number of fluorine atoms in the polymer is 1 : 2.

i) Draw a portion of the Teflon structure with 8 carbon atoms.

ii) Draw the structure of the monomer used in producing Teflon.

25 It is the job of a materials scientist to select the best material for a particular purpose. The table below gives information about the properties of five polymers.

Polymer	Density (kg m^{-3})	Tensile strength (MPa)	Maximum operating temperature (°C)	Dissolves in organic solvents?	
				below 80 °C	above 80 °C
Polystyrene	1 050	40	65	Yes	Yes
Low density polythene (LDPE)	920	15	85	No	Yes
High density polythene (HDPE)	960	29	120	No	Yes
Polypropene	900	35	160	No	No
Polychloroethene	1 390	60	60	Yes	Yes

a) Give the definition of the term 'polymer'.
A polymer is a compound which consists of very large molecules formed by joining together many small molecules repeatedly.

b) Why is HDPE more suitable than LDPE for making milk crates?
HDPE has a higher tensile strength. / HDPE is stronger.

c) Use the information in the table to suggest ONE use for LDPE.
Film and sheet for packaging / wrapping / grocery bag / squeezable bottle

d) i) Which polymer in the table would be best for making a pipe to carry oil at a temperature of 120 °C?
Polypropene

ii) Give TWO reasons for your choice.
Reason 1: Does not dissolve in organic solvents.
Reason 2: Maximum operating temperature is above 120 °C.

(AQA Advanced Subsidiary GCE, Applied Science, Unit 5, Jan. 2010, 3)

Topic Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Multiple choice questions

1 Which of the following substances, obtained from the fractional distillation of crude oil, has the lowest boiling temperature?

(A) Refinery gas
B Kerosene
C Diesel oil
D Lubricating oil

Refinery gas is a gas while the others are liquids at room temperature and pressure. Thus, refinery gas has the lowest boiling point.

(Edexcel Advanced Subsidiary GCE, Unit 1, Jun. 2009, 10)

2 Which of the following is a greenhouse gas?

A Argon
B Nitrogen
C Oxygen
(D) Water vapour

(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2011, 5)

3 Ethane reacts with chlorine when the substances are exposed to UV radiation. The role of the UV radiation in the reaction is to

(A) break the Cl–Cl bond forming Cl• free radicals.
B break the Cl–Cl bond forming Cl^+ and Cl^- ions.
C break the C–C bond in ethane forming CH_3• free radicals.
D break a C–H bond in ethane forming C_2H_5• free radicals.

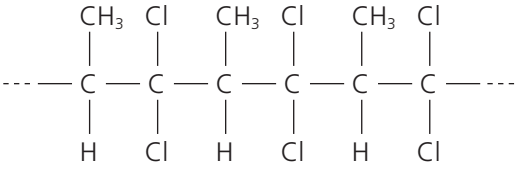
(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2011, 8(b))

4 Propene reacts with bromine dissolved in an organic solvent to give

A 1-bromopropane.
B 2-bromopropane.
(C) 1,2-dibromopropane.
D 1-bromopropane and 2-bromopropane.

(HKCEE, Paper 2, 2011, 13)

5 A portion of the structure of an addition polymer X is shown below:



Which of the following is the systematic name of the monomer of X based on the given structure?

A 1,1-dichloro-2-methylethene
(B) 1,1-dichloropropene
C 1,2-dichloropropene
D 3,3-dichloropropene

(HKDSE, Paper 1A, 2013, 14)

6 Which of the following measures can help reduce the level of carbon monoxide at the road side?

(1) Using liquefied petroleum gas as fuel for motor vehicles
(2) Installing catalytic converters for motor vehicles
(3) Installing electrostatic precipitators for motor vehicles

(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKCEE, Paper 2, 2009, 26)

7 The exhaust pipe of a petrol-driven car will corrode easily because

(1) there are nitrogen oxides passing through the exhaust pipe.
(2) there are unburnt hydrocarbons passing through the exhaust pipe.
(3) the exhaust pipe usually becomes hot when the car engine is running.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2011, 42)

8 When chlorine reacts with methane under sunlight, which of the following compounds can be formed?

(1) Chloromethane
(2) Dichloromethane
(3) Hydrogen chloride

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKCEE, Paper 2, 2008, 24)

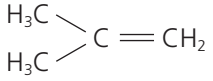
9 Which of the following statements concerning cracking and fractional distillation in petrochemical industry is / are correct?

(1) Both processes involve heating.
(2) Both processes are chemical changes.
(3) Both processes produce extra alkenes.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2011, 22)

10 Consider the following organic compound:



Which of the following statements about this compound is / are correct?

(1) Its systematic name is 1,1-dimethylethene.
(2) It can decolourize an acidified solution of potassium permanganate.
(3) It is the monomer of Perspex.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKDSE, Practice paper 1A, 2012, 21)

Directions : *Each question (Questions 11–15) consists of two separate statements. Decide whether each of the two statements is true or false; if both are true, then decide whether or not the second statement is a correct explanation of the first statement. Then select one option from A to D according to the following table :*

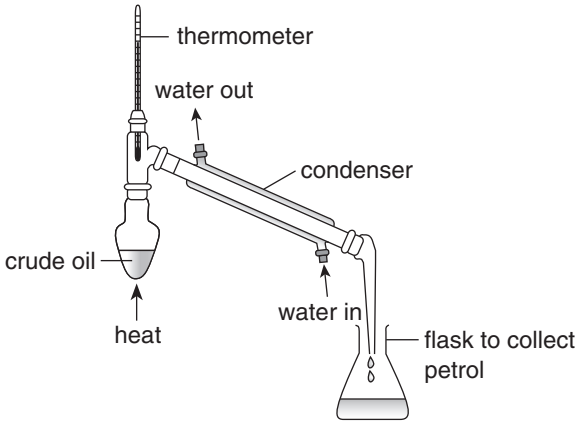
A Both statements are true and the 2nd statement is a correct explanation of the 1st statement.
B Both statements are true but the 2nd statement is NOT a correct explanation of the 1st statement.
C The 1st statement is false but the 2nd statement is true.
D Both statements are false.

	1st statement	2nd statement	
11	Burning fossil fuels can cause acid rain.	Burning fossil fuels produces carbon dioxide. *(HKDSE, Paper 1A, 2012, 24)*	B
12	Unpolluted rainwater can erode limestone.	Carbon dioxide in air dissolves in unpolluted rainwater to form carbonic acid. *(HKCEE, Paper 2, 2011, 28)*	A
13	Fractional distillation can convert large alkane molecules to smaller alkane molecules and alkene molecules.	Fractional distillation involves breaking and forming of covalent bonds. *(HKCEE, Paper 2, 2008, 49)*	D
14	The boiling point of butane is higher than that of methane.	The van der Waals' forces between butane molecules are stronger than the forces between methane molecules. *(HKCEE, Paper 2, 2008, 29)*	A
15	Propane can change acidified potassium permanganate solution from purple to colourless.	Substitution reaction occurs when propane is added to acidified potassium permanganate solution. *(HKCEE, Paper 2, 2011, 50)*	D

Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

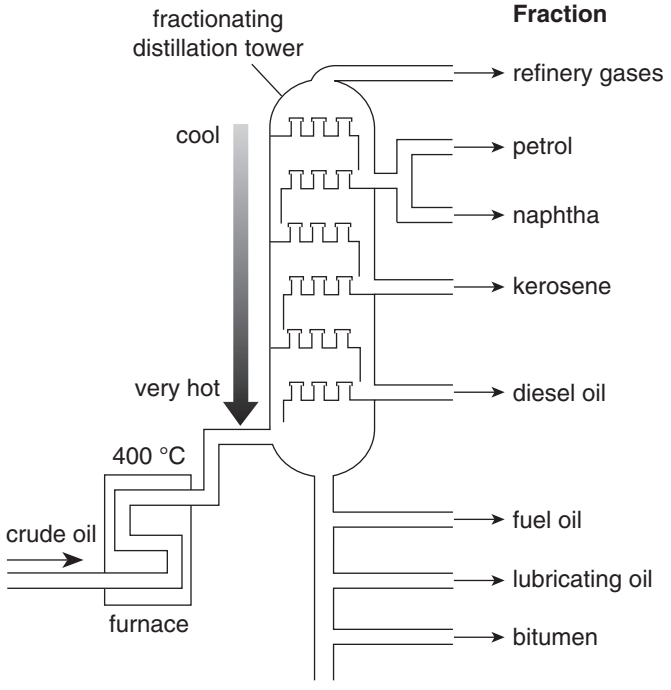
16 Crude oil is a mixture of different hydrocarbons. Petrol can be separated from this mixture using the process of distillation. The mixture is heated and the petrol collected at a temperature of 40–100 °C.



a) Explain how this process of distillation works to separate petrol from crude oil.

b) In an oil refinery the distillation process has to be scaled up.

A fractional distillation tower is used to separate the crude oil into its different components.



One advantage of this process is that several other useful chemicals can be separated at the same time.

Give TWO other advantages of distilling crude oil in this way.

c) The Health and Safety Executive has set-up regulations which aim to protect people at work.

i) Suggest a precaution that workers at the oil refinery should take to keep themselves safe.

ii) Suggest a precaution that workers at the oil refinery should take that also keeps others safe.

(OCR GCSE 21st Century Science (Higher Tier), Additional Applied Science A, A335, Jan. 2009, 4)

17 To reduce air pollution caused by vehicles, several measures have been adopted in recent years.

a) Many taxis and mini-buses have switched from using diesel to liquefied petroleum gas (LPG) as fuel.

i) Give the name of a compound which is a major component of LPG.

ii) Why is LPG considered to be a 'cleaner' fuel than diesel?

b) Catalytic converters have been installed in most petrol-driven vehicles.

i) State TWO functions of catalytic converters.

ii) State ONE harmless product emitted from catalytic converters.

c) Some regions supply ultra low sulphur diesel (ULSD) for diesel vehicles. Explain how this measure reduces air pollution.

(HKCEE, Paper 1, 2011, 6)

16 a) The petrol vaporizes and then the vapour condenses on the condenser and is collected in the conical flask finally. Others are left in the original flask.

b) The process is continuous / constantly feed in crude oil and collect the petrol. This process involves automation. Less labour is required.
The huge fractionating tower can process large amount of petroleum at the same time which facilitates mass production.

c) i) Wear safety glasses / face masks / other correct precautions.

ii) No smoking / no naked flame / report spills / other correct precautions.

18 There is much international concern that an increase in atmospheric concentrations of carbon dioxide and methane may lead to global warming and climate change.

Refer to page T54.

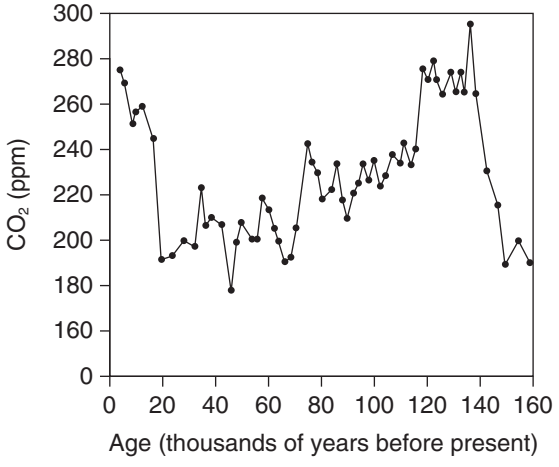
a) Carbon dioxide and methane are both greenhouse gases.

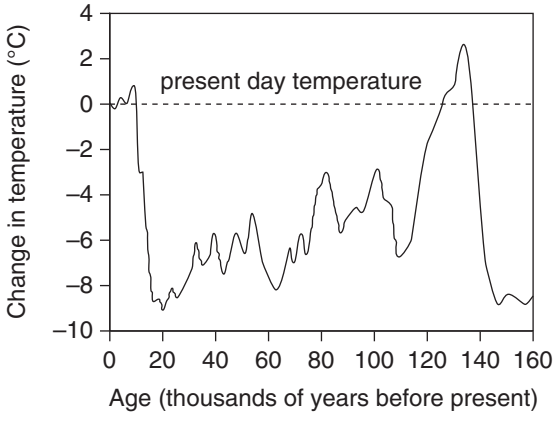
Some scientists are more concerned about carbon dioxide as a greenhouse gas than methane. Suggest why.

b) Describe ways that research chemists are trying to minimize climate change resulting from global warming caused by the release of greenhouse gases.

c) Research scientists working in the Antarctic have measured the concentration of carbon dioxide in the ice. This study has allowed the scientists to estimate the atmospheric concentration of carbon dioxide over many thousands of years.

The graphs below show these atmospheric concentrations and the corresponding average surface temperature.





Do the graphs provide reliable evidence that an increase in atmospheric carbon dioxide concentration will result in global warming? Explain your answer.

(OCR Advanced Subsidiary GCE, Chem. A, F322, Jan. 2010, 7)

19 Cetane ($C_{16}H_{34}$) is major component of diesel fuel.

a) Write an equation to show the complete combustion of cetane.

b) Cetane has a melting point of 18 °C and a boiling point of 287 °C. In polar regions vehicles that use diesel fuel may have ignition problems. Suggest ONE possible cause of this problem with the diesel fuel.

c) The pollutant gases NO and NO_2 are sometimes present in the exhaust gases of vehicles that use petrol fuel.

i) Write an equation to show how NO is formed and give a condition needed for its formation.

ii) Write an equation to show how NO is removed from the exhaust gases in a catalytic converter. Identify a catalyst used in the converter.

iii) Deduce an equation to show how NO_2 reacts with water and oxygen to form nitric acid (HNO_3).

d) Cetane ($C_{16}H_{34}$) can be cracked to produce hexane, butene and ethene.

i) Write an equation to show how one molecule of cetane can be cracked to form hexane, butene and ethene.

ii) State ONE type of useful solid material that could be formed from alkenes.

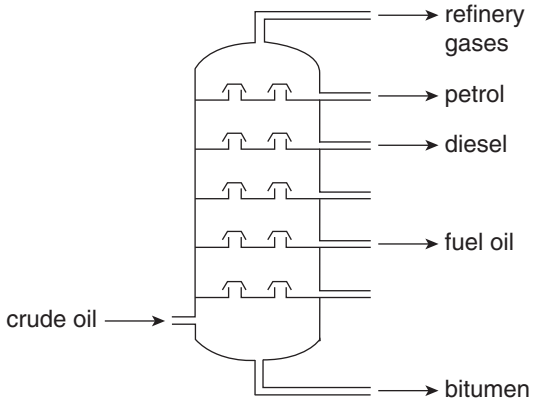
(AQA Advanced Subsidiary GCE, Unit 1, Jan. 2011, 4)

19 a) $2C_{16}H_{34}(l) + 49O_2(g) \longrightarrow 32CO_2(g) + 34H_2O(l)$
b) The fuel solidifies.
c) i) Equation: $N_2(g) + O_2(g) \longrightarrow 2NO(g)$
Condition: Very high temperature
ii) Equation: $2NO(g) + 2CO(g) \longrightarrow N_2(g) + 2CO_2(g)$
Catalyst used: platinum / rhodium / palladium
iii) $4NO_2(g) + 2H_2O(l) + O_2(g) \longrightarrow 4HNO_3(aq)$
d) i) $C_{16}H_{34}(l) \longrightarrow C_6H_{14}(l) + 2C_4H_8(g) + C_2H_4(g)$
or $C_{16}H_{34}(l) \longrightarrow C_6H_{14}(l) + C_4H_8(g) + 3C_2H_4(g)$
ii) Synthetic polymers

20 Crude oil is a fossil fuel.

a) Crude oil is separated into useful substances by fractional distillation.

Look at the diagram. It shows a fractionating column.



The refinery gases come out of the top of the fractionating column. Explain why. Use ideas about boiling points.

Refinery gases have the lowest boiling point range.

b) Look at the table.

It shows the amount of some fractions made from 100 tonnes of crude oil. It also shows the amount of each fraction needed for everyday use.

Fraction	Amount made in tonnes (supply)	Amount needed in tonnes (demand)
Refinery gases	5	10
Petrol	10	25
Diesel	15	15
Fuel oil	40	30

i) For which fraction does the supply match the demand?

Diesel

ii) Some fractions from crude oil are cracked to make petrol.

Cracking is used to match supply with demand. Explain how. Use the information in the table to help you.

Cracking breaks down larger hydrocarbon molecules with heat or catalyst to produce smaller hydrocarbon molecules.

Cracking converts fractions in excess to those in demand.

c) The petrol fraction contains octane, C_8H_{18}.

Octane contains only carbon and hydrogen atoms.

Octane is a hydrocarbon. Explain why.

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 1, Jun. 2011, 1)

21 Olympic torches have used a variety of fuels since the first Olympic games in 776 B.C. Propane gas was the fuel chosen for the Beijing Olympics in 2008.

One problem with using propane is that the flame is not very visible. In 1996, the fuel used was a mixture of propene, C_3H_6, and propane gases which made the Olympic flame brighter with a more visible yellow flame.

a) Draw the skeletal formulae for propane and propene in the boxes below.

Propane	Propene
	

b) The percentage by mass of carbon in propane is 82%.

Calculate the percentage by mass of carbon in propene to show that it is larger than 82%.

c) Use the information from (b) to suggest why the 1996 torch produced a much more visible yellow flame compared to the Beijing torch.

(OCR Advanced Subsidiary GCE, Chem. B (Salters), F331, Jun. 2011, 1(b))

22 Use electron diagrams to illustrate, step by step, how CH_4 reacts with Br_2 under sunlight to form CH_3Br. (Show electrons in the *outermost shells* only.)

(HKDSE, Paper 1B, 2012, 15)

21 b) Percentage by mass of carbon in propene

$= \frac{36.0}{42.0} \times 100\%$

$= 85.7\%$

c) Propene has a higher carbon to hydrogen ratio. It is more difficult for propene to undergo complete combustion. Carbon particles are produced in the process.

23 In 1860, Michael Faraday gave a series of lectures in London entitled 'The Chemical History of a Candle'. Paraffin wax was, and still is, one of the materials used to make candles.

a) Paraffin wax is a mixture of aliphatic hydrocarbons with more than 20 carbon atoms per molecule. Hydrocarbons of this length can be obtained from crude oil.

i) Explain the term 'aliphatic'. Hydrocarbon that does not contain benzene ring / aromatic structure.

ii) What is the name of the process used to separate hydrocarbons of different chain lengths from crude oil?
Fractional distillation

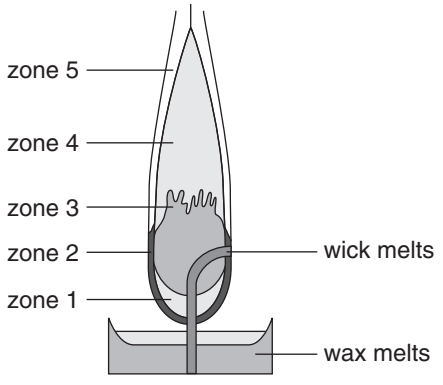
iii) Paraffin wax can be represented by the formula $C_{25}H_{52}$.

Balance the following equation to represent the complete combustion of $C_{25}H_{52}$.

$C_{25}H_{52} + \underline{38}O_2 \longrightarrow 25CO_2 + 26H_2O$

b) The diagram below outlines some areas or zones in a candle flame.

Zone 2 represents the hottest part of the flame and equation in (a)(iii) represents the reaction taking place in this zone.



i) Zone 1 is much cooler than the flame. If a glass tube is placed in the centre of this zone, the vapour coming out of the end of the tube can be ignited.

Suggest a substance, present in the vapour, that could be ignited.
Paraffin wax / $C_{25}H_{52}$

ii) Incomplete combustion occurs in Zone 4. Suggest TWO products of incomplete combustion present in Zone 4.
Any two of the following:
Carbon monoxide / carbon / soot

c) Oil refineries sometimes 'crack' paraffin waxes to produce more useful products.

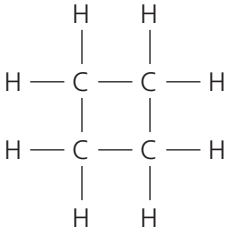
The products formed from the cracking process always include an unsaturated compound such as the one shown in the diagram below.



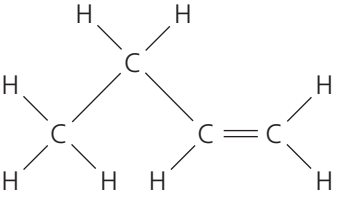
What is the molecular formula of the unsaturated molecule shown above? C_3H_6

(OCR Advanced Subsidiary GCE, Chem. B (Salters), F331, Jan. 2010, 3(a)–(c))

24 A and B are two hydrocarbons with the molecular formula C_4H_8. Their structures are:



A

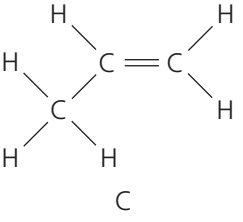


B

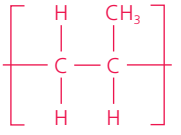
a) Describe a chemical test to distinguish between hydrocarbons A and B. Give the result you would expect for each hydrocarbon.
Refer to the next page.

b) To which homologous series does B belong?
Alkenes

c) Hydrocarbon C belongs to the same homologous series as B. It has the structure:



C



C forms an addition polymer. Draw the repeating unit of this polymer.

d) Give the name of the addition polymer formed by C. Polypropene

(Edexcel IGCSE (Higher Tier), Chemistry, Paper 2H, Jun. 2010, 7)

25 This question is about ethene and related compounds.

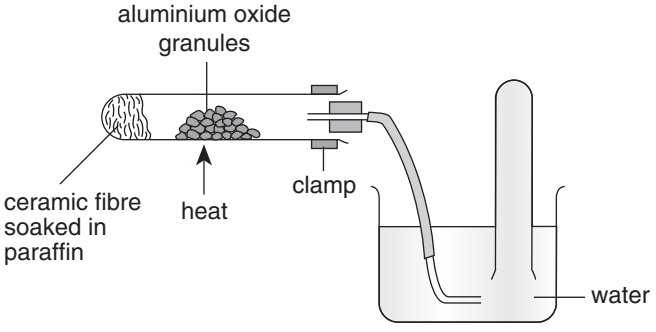
Refer to page T54.

a) One way to manufacture ethene is by cracking hydrocarbon molecules such as liquid paraffin.

i) Name a raw material from which liquid paraffin can be obtained.

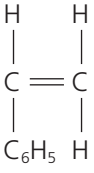
ii) Describe what is meant by 'cracking'.

iii) It was proposed to set up the apparatus below on a laboratory bench, in order to crack paraffin.



State TWO of the risks of using the apparatus in this way and suggest how you would amend the set-up to minimize each risk.

b) The formula of the alkene phenylethene, often called styrene, is shown below. It can be used to make the polymer polystyrene.



phenylethene

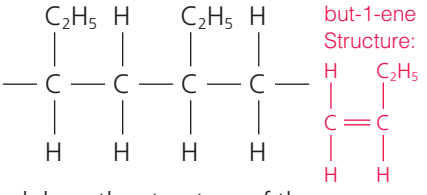
Draw a section of the polystyrene polymer chain formed from two monomer units.

(Edexcel Advanced Subsidiary GCE, Unit 1, Jun. 2010, 18(a), (d))

24 a) Any one of the following:
- Mix each hydrocarbon with bromine dissolved in an organic solvent separately.
  The orange solution of bromine becomes colourless rapidly when mixed with hydrocarbon B.
  The orange solution of bromine becomes colourless slowly when mixed with hydrocarbon A under sunlight.
- Mix each hydrocarbon with cold acidified dilute potassium permanganate solution separately.
  The purple solution of potassium permanganate becomes colourless rapidly when mixed with hydrocarbon B.
  There is no observable change for hydrocarbon A.

26 Plastics contain polymer molecules. The disposal of waste plastics is causing many environmental problems. In the middle of the Pacific Ocean, there is a huge area of sea water contaminated with very small pieces of plastic waste.

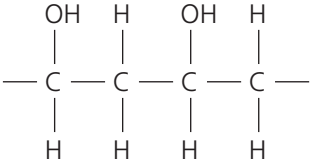
a) A short section of one of the polymers found in the Pacific Ocean is shown below.



Name of monomer: but-1-ene
Structure:

Name and draw the structure of the monomer used to produce this polymer.

b) The short section of polyethenol is shown below.



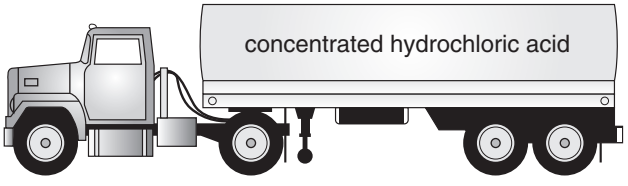
Large quantities of polythene and polyethenol are disposed of each year.

Polythene is found as a waste plastic in the Pacific Ocean but polyethenol is not because it slowly dissolves in water. The polymer contains many –OH groups. It can form hydrogen bonds with water.

Suggest why polyethenol dissolves in water.

(OCR Advanced Subsidiary GCE, Chem. A, F322, Jun. 2011, 6(a))

27 The diagram below shows a truck with a storage tank for transporting concentrated hydrochloric acid.



a) Suggest a hazard warning label that should be posted on the storage tank.

b) The storage tank is made of steel and the inner wall has a lining of polythene.

i) Draw the structural formula of polythene.

ii) Explain the function of the lining of polythene in terms of the chemistry concept involved.

(HKCEE, Paper 1, 2009, 4)

28 Petrol contains saturated hydrocarbons. Some of the molecules in petrol have the molecular formula C_8H_{18} and are referred to as octanes. These octanes can be obtained from crude oil by fractional distillation and by cracking suitable heavier fractions.

Petrol burns completely in a plentiful supply of air but can undergo incomplete combustion in a car engine.
Refer to page T55.

a) State the meaning of both the words 'saturated' and 'hydrocarbon' as applied to the term 'saturated hydrocarbon'.

Name the homologous series to which C_8H_{18} belongs.

b) Outline the essential features of the fractional distillation of crude oil that enable the crude oil to be separated into fractions.

c) C_8H_{18} is obtained by the catalytic cracking of suitable heavy fractions.

i) State what is meant by the term 'cracking'.

ii) Write an equation to show how one molecule of $C_{14}H_{30}$ is cracked to form one molecule of C_8H_{18} and one molecule of another hydrocarbon.

iii) Explain why oil companies need to crack 'suitable heavy fractions'.

d) Write an equation for the incomplete combustion of C_8H_{18} to form carbon monoxide and water only.

e) A catalytic converter is used to remove carbon monoxide from the exhaust gas in a car.

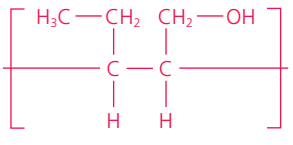
i) Identify a catalyst used in the catalytic converter.

ii) Write an equation to show how carbon monoxide is removed in a catalytic converter.

f) State why the water produced in the exhaust gas may contribute to global warming.

(AQA Advanced Subsidiary GCE, Unit 1, Jan. 2009, 6)

30 d) i)

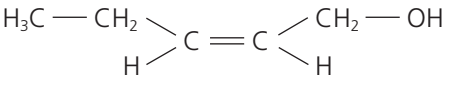


29 Briefly describe how polypropene can be produced from naphtha.

(Communication mark will be awarded.)

(HKDSE, Paper 1B, 2013, 6)

30 'Violet oil' is sometimes used in aromatherapy treatments for its mild pain-killing properties. The oil has a strong 'leafy' odour due partly to the presence of compound A.



compound A

a) Name the functional groups present in compound A.
Hydroxyl group
Carbon-carbon double bond

b) Give the molecular formula of compound A.
$C_5H_{10}O$

c) A student reacts compound A with bromine dissolved in an organic solvent.

i) Describe the colour change the student would see when this reaction takes place.
The orange solution of bromine becomes colourless.

ii) Write the equation for the reaction between compound A and bromine.
$C_5H_{10}O + Br_2 \longrightarrow C_5H_{10}OBr_2$

d) Compound A can also be used to produce an addition polymer that is soluble in water.

i) Draw a diagram to show the repeating unit of the polymer formed from compound A.

ii) The polymer formed from compound A is soluble in water.

Explain this solubility in terms of the molecular structure of the polymer and the formation of hydrogen bonds.

(OCR Advanced Subsidiary GCE, Chem. B (Salters), F332, Jan. 2012, 2(a), (c), (f))

30 d) ii) The hydrogen atoms of the –OH groups of the polymer have strong partial positive charges.
Hydrogen bonds form between these hydrogen atoms and the lone pairs on the highly electronegative oxygen atoms of water molecules.
OR
The hydrogen atoms of the water molecules have strong partial positive charges.
Hydrogen bonds form between these hydrogen atoms and the lone pairs on the highly electronegative oxygen atoms of the –OH groups of the polymer.
The polymer is soluble in water because it can form hydrogen bonds with water.